Airspace Reroute Simulation Under Threat

AI tools are allowed and encouraged. Use any language or libraries you like.

Scenario

Aircrafts are flying through a shared block of airspace. A potentially unsafe object is flying in a straight line at constant speed toward a fixed ground target. The target is not any flying aircraft. Your task is the following: define and analyze the problem, build the data, and then build a simulation that keeps every aircraft as safe as possible for the next 20 **minutes**. Assume that at all times, aircrafts know the position of the unsafe flying object, as well as the positions of all other aircrafts.

The ask

1. **Analyze the scenario.** You are given a snapshot of aircraft at t = 0 (position, altitude, velocity) plus the hostile object and the target. Download the given files.
2. **Simulate the next 20 minutes** (1200 s). Keep every aircraft safe (your own definition of safe) from the threat, the target, and each other, while each one continues toward its destination. An aircraft's destination is where its original straight-line path would take it. You may not teleport or remove aircraft.

What is provided

- **Input specs:** t=0 snapshot (id, location, velocity, altitude) of the aircrafts and hostile object. And the area of the airspace.

  - *Flights.csv - Data about aircrafts at t=0*

Column	Description
flight_id	Unique aircraft identifier
x_m	East-west position (m)
y_m	North-south position (m)
alt_m	Altitude (m)
vx_mps	East-west velocity (m/s)
vy_mps	North-south velocity (m/s)
vz_mps	Climb/descent rate (m/s)

  - *scenario.json - Data on hostile object (t=0 location, velocity, altitude), corridor area (x, y, z bounding box coordinates of airspace), and location of target*

What is up to you

- **The safety rules:** danger zone sizes around the target and hostile object (2D or 3D), minimum separation between aircraft, and any buffers.

(Hint) This should include: Safety buffer around the hostile threat and target (minimum distance zones that the aircrafts should avoid, can be several km radius around the threat), safe flying buffer between the aircrafts, and any other "safety" specs do you think are necessary.

- **The aircraft specs:** speed range, turn rate, climb rate, and anything else you think matters.
- **The approach:** which maneuvers you allow for aircraft (left, right, up, down etc.), how you detect conflicts, and how far ahead you look.
- **Quality:** how you define "good" (deviation, delay, rejoining, workload) and how you trade it against safety.
- **Validation:** how you convince us the result is safe.

What to deliver

- [The data] A trajectory file for all aircraft over time.
- [The rules] A config file (for example `rules.json`) holding every rule and limit you chose, used by both your planner and your checker.
- Evidence that your result meets your own rules.
- A short plot or animation (nice to have).
- A README (one page or less) that **starts with your scenario design and the rules you chose and why**, then covers your approach, assumptions, and known limits.

If something is ambiguous, state your assumption and move on. We care about your reasoning as much as the output.

Quick sanity facts (optional help)

- A jet at 230 m/s covers 230 x 1200 = 276 km in 15 minutes, The hostile object at 120 m/s covers 144 km.
- At 230 m/s and 3 deg/s, the tightest turn radius is about 4.4 km.
- 300 m of altitude change takes at least 20 s at 15 m/s.